

YVES SHABO NKULU-ILY

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ACADEMIC CURRICULUM VITAE | RESEARCH, TEACHING, AND SERVICE

Research Profile

Interdisciplinary learning design and educational technology researcher-practitioner working at the intersection of instructional design, distance education, XR/AI, haptics, simulation-based learning, and medical education. Ph.D. Candidate in Education at the University of Kentucky and disclosed originator of a University-reviewed invention report for an immersive 3-D augmented-reality orthopedic learning environment.

Research Interests

- XR/AI and haptics for medical and surgical education
- Simulation-based learning and realistic decision-making
- Feedback design, cognitive fit, and instructional alignment
- Distance education and e-learning curriculum implementation
- Learning design for resource-constrained and distributed environments
- Educational technology adoption, technical training, and capacity building

Education

University of Kentucky | Lexington, KY

Aug 2017–Present

Ph.D. Candidate in Education, Teaching and Learning / Learning Design & Technology

- Graduate Certificate, Distance Education, 2020
- Graduate Certificate, Research Methods in Education, 2019

Institute of Statistics of Lubumbashi | Lubumbashi, Democratic Republic of the Congo

Sep 1995–Oct 2000

Licentiate Degree in Statistics, computing focus

Sitec Training College Ltd | United Kingdom

Jan 2004–Jun 2004

OCR Certificate for IT Practitioners, Level II; Cisco Networking Academy IT Essentials I embedded

Academic and Research Experience

University of Kentucky | Lexington, KY

Aug 2017–May 2021

Teaching Assistant

Supported EDC 317, Introduction to Instructional Media, a foundational 1-credit course for Semester 6 pre-service teachers in Elementary Education, Middle Level Education, and Learning and Behavior Disorders.

Guided candidates in selecting, operating, producing, and integrating analog and digital media with instructional goals, using multimedia design frameworks to reduce cognitive load.

Supported hands-on work with video-editing, presentation, image-manipulation, and learning-management tools, including Canvas where used in the course.

Reinforced Universal Design for Learning (UDL), concise instructional explanation, and intentional multimedia theory through production of 2- to 3.5-minute digital media concepts and other multimedia documents.

Independent / Freelance | Lubumbashi, Democratic Republic of the Congo / Remote

Feb 2009–Present

Proposal Writer / Research Contributor

- Developed and coordinated proposal materials for public health, education, new media, technical workshops, and research initiatives.
- Monitored requests for proposals, identified opportunities, and coordinated inputs and deadlines.

Project Designer

- Helped shape and implement a French Cooperation-funded higher-education computing initiative.
- Selected learning materials and technology-enabled methods to strengthen the instructional environment.
- Represented the project in partnership discussions and helped secure partner participation.
- Designed and developed the ESIS website with online applications for new students.
- Supported training that reached more than 1,000 IT technicians.

Peer-Reviewed Publications

- Nkulu-Ily, Y. S., Gantwerker, E. A., Vaughan, N., Cheng, Q., & Dhanda, J. (2026). Simulating realistic surgeon decisions and actions in immersive medical distance education with XR and AI. *Interactive Learning Environments*, 34(4), 2223–2268. DOI: 10.1080/10494820.2025.2542893.
- Nkulu-Ily, Y. S. (2023). Implementation of e-learning curriculum in higher education. *European Journal of Open, Distance and E-Learning*, 25(1), 62–73. DOI: 10.2478/eurodl-2023-0005.
- Nkulu-Ily, Y. S. (2023, May). Combining XR and AI for integrating the best pedagogical approach to providing feedback in surgical medical distance education. In *International Conference on Intelligent Tutoring Systems* (pp. 452–466). Springer Nature Switzerland. DOI: 10.1007/978-3-031-32883-1_41.

Preprints, Manuscripts, and Public Scholarship

- Yves S. Nkulu-Ily (2026). Bridging Distance with Sensation: Crafting Immersive Haptic Feedback for Enhanced Orthopedic Education in XR/AI [Preprint]. DOI: 10.2139/ssrn.6568980.
- Yves S. Nkulu-Ily (2026). Operationalizing alignment and cognitive fit in AI-driven WebXR systems for orthopedic surgical distance education [Manuscript submitted for publication]. *Interactive Learning Environments*.
- Yves S. Nkulu-Ily (in preparation). Operationalizing a candidate optimal challenge point in AI-supported WebXR orthopedic surgical distance education: Phase 0 construct specification and a Phase 1 calibration and randomized counterbalanced crossover pilot [Manuscript submitted for publication].
- Yves S. Nkulu-Ily (2015). Proposed Orthopedics and Traumatology Project in Lubumbashi, Democratic Republic of Congo. Unpublished manuscript.
- Yves S. Nkulu-Ily, & Witry, J. (2018). Elaboration of a structural equation model based on individuals' vocational interests, personality, and cognitive ability. University of Kentucky. Unpublished manuscript.
- Yves S. Nkulu-Ily (2026, May 5). Adaptive XR & AI for total knee replacement (TKR) training. *The Science Matters*.

Innovation and Intellectual Property

- Originator, Invention Report 2541, “The Design of an Internet-Based Immersive 3-D Digital Orthopedics Learning Environment Using Augmented Reality.” University of Kentucky Intellectual Property Committee decision, December 15, 2020.
- IPC determination: University legal ownership interest; Office of Technology Commercialization directed to investigate appropriate protection on behalf of the University of Kentucky Research Foundation.

Professional Experience

Catholic Relief Services | Lubumbashi, Democratic Republic of the Congo**Aug 2008–Feb 2009***MIT*

- Supported network infrastructure, troubleshooting, backups, archiving, and disaster-recovery activities.
- Identified system requirements and analyzed business rules and process needs.
- Trained 15 staff using e-learning and team-based learning methods.

SOCODAM s.p.r.l. | Lubumbashi, Democratic Republic of the Congo**Jan 2007–Jul 2008***IT Manager*

- Designed IT network infrastructure and trained two dozen staff.
- Led IT training and program-improvement activities and supported IT operations and financial administration.
- Reorganized filing systems to improve office efficiency and customer service.

Computer Consultant

- Contributed technical expertise to consulting engagements and developed an application supporting computerized delivery of information, advice, and guidance.

Teaching and Training

- Teaching Assistant, University of Kentucky, 2017–2021.
- Technical training for more than 1,000 IT technicians through the ESIS initiative.
- Staff training for 15 CRS employees using e-learning and team-based learning methods.
- IT infrastructure and staff training for two dozen SOCODAM employees.

Service and Community Engagement

- Co-Founder / Volunteer Supervisor, NGO AKILIMALI, Lubumbashi, July 2003–Present.
- Volunteer involvement with Oxfam Great Britain and Amnesty International Belgium.

Skills and Languages

- Learning Design; Instructional Design; Educational Technology; E-Learning; Distance Education; XR/AI and Haptics; Medical Education; Simulation-Based Learning; Research Methods.
- Technical Training; Proposal and Grant Writing; Microsoft Windows; Microsoft Office Suite; Network Support; Software Installation; Technical Troubleshooting; Backups; Archiving; Disaster Recovery; Knowledge of Programming.
- English (fluent/bilingual); French (native/bilingual); Swahili (conversational).